Windows Server 2022 Domain Infrastructure Setup and Configuration

Contents

1 Introduction

1.1 Purpose of the Report

The purpose of this report is to document the installation, configuration, and implementation of a Windows Server 2022 environment, including the setup of Active Directory Domain Services (AD DS), Group Policy Management, and the integration of client computers into the domain. This report serves as a technical guide and record of the steps followed, ensuring a structured approach to building a domain-based infrastructure. It also provides a reference for troubleshooting, future improvements, and replicating the setup in similar environments.

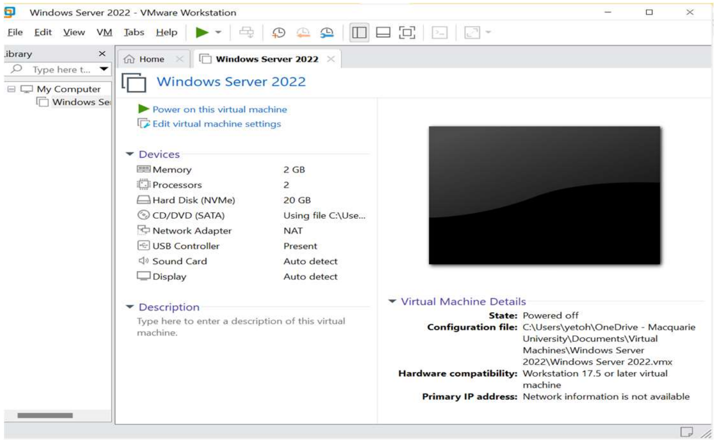
2 Lab Setup

For this lab environment, VMware Workstation is used to host the virtualized infrastructure. The specific configuration for the primary server is as follows:

Virtualization Platform: VMware Workstation 17.5 or later

Windows Server 2022 Virtual Machine Configuration:

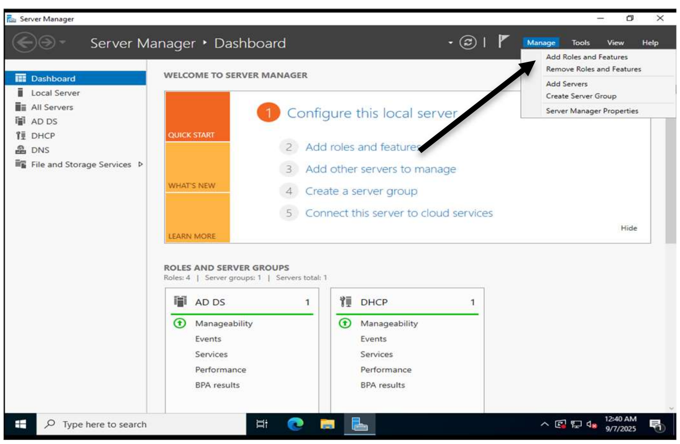
- **Memory: 2 GB** RAM allocated. This meets the minimum requirement for Windows Server 2022 and is suitable for a domain controller with light loads in a lab environment.
- **Processors: 2** virtual CPU cores allocated. This provides sufficient processing power for core services like Active Directory and DNS.
- **Hard Disk: 20 GB** NVMe virtual hard disk. The NVMe controller provides better disk I/O performance. This size is adequate for the OS and core server roles.
- **CD/DVD (SATA):** Configured to use an ISO image file for installation.
- **Network Adapter:** Connected to the **NAT** network.
    - *Why NAT?* This choice provides the virtual machine with internet access (through the host's network connection) for updates and licensing, while also protecting it from external networks. It is the simplest way to get a lab environment online.
- **USB Controller, Sound Card, Display:** Set to Auto Detect, as these are not critical for a server core infrastructure lab.

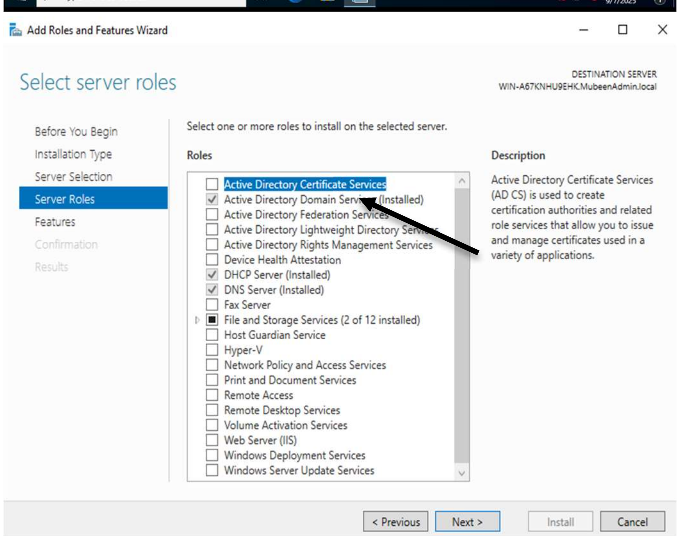


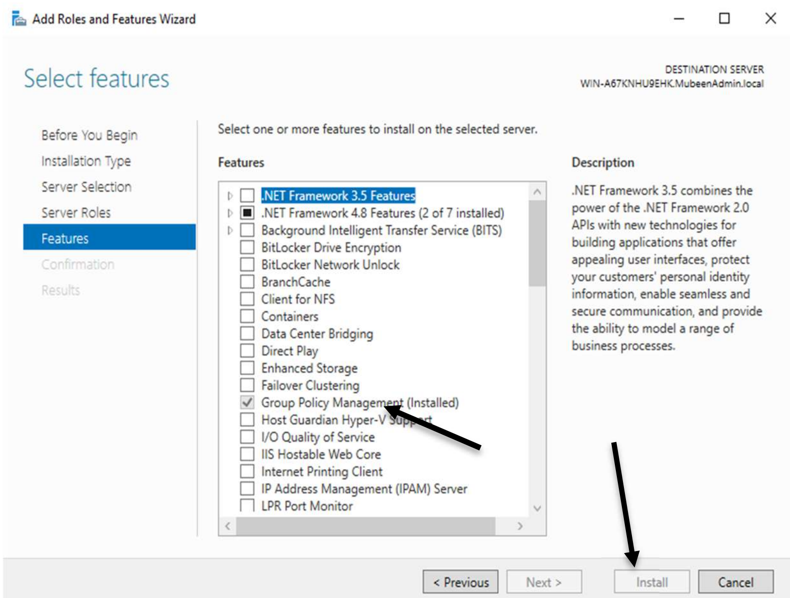
3. Windows Server 2022 Installation

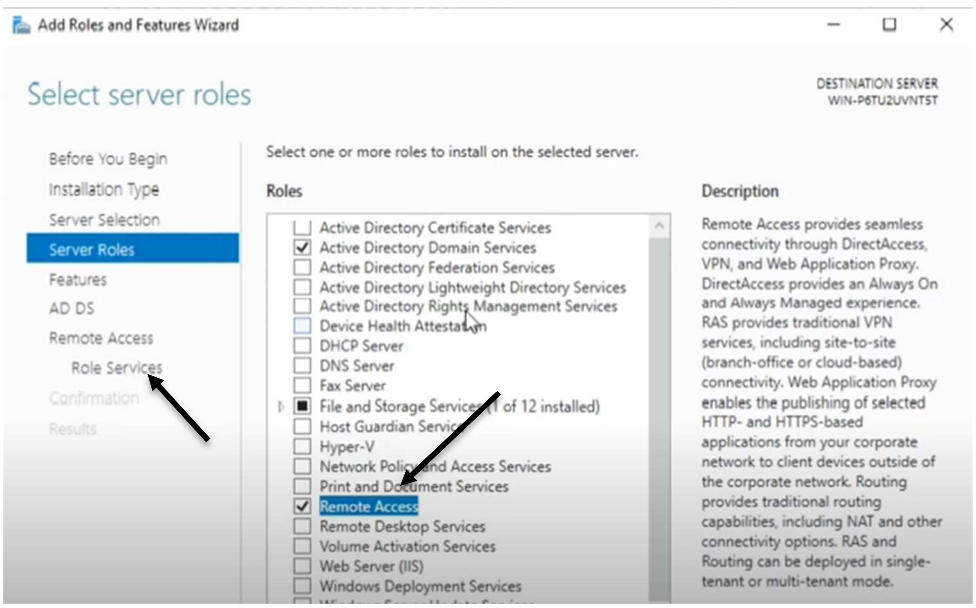
3.1 Installing Active Directory Domain Services (ADDS)

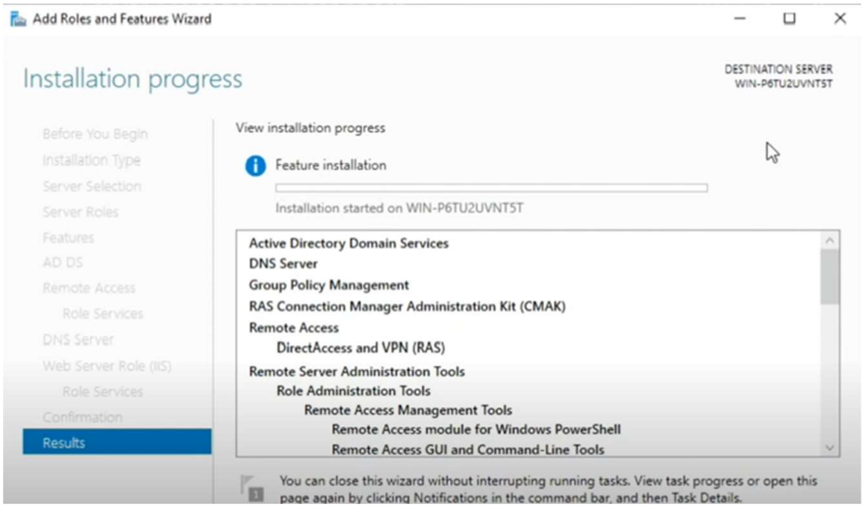
Active Directory Domain Services



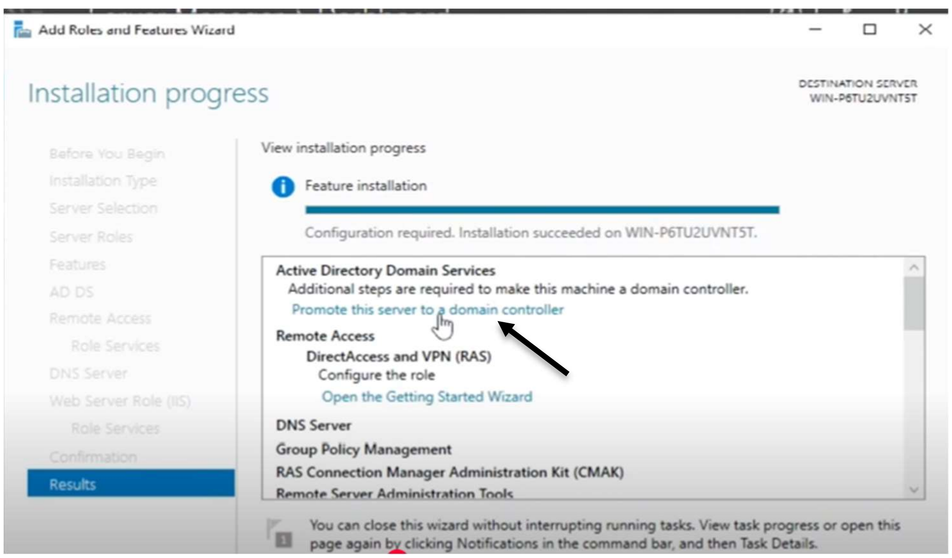


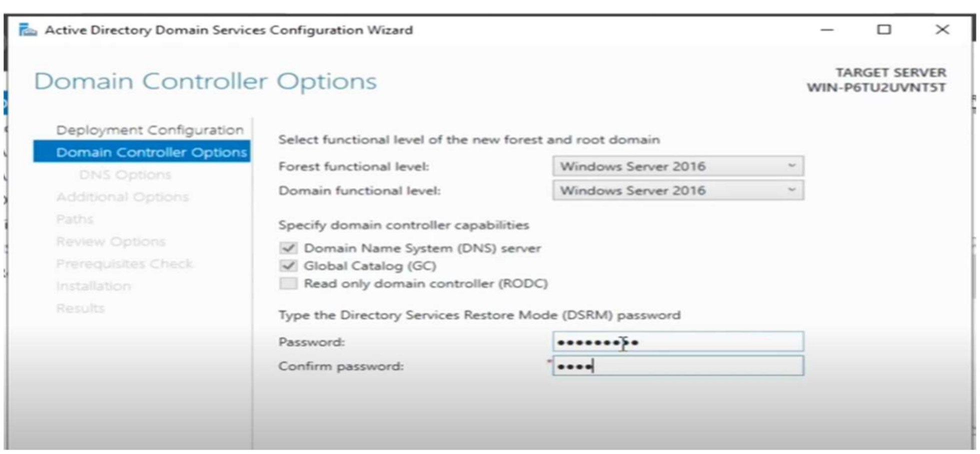

Add Roles and Features Wizard
Select features
DESTINATION SERVER
WIN-A67KNHU9EHK.MubeenAdmin.local
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more features to install on the selected server.
Features
.NET Framework 3.5 Features
.NET Framework 4.8 Features (2 of 7 installed)
Background Intelligent Transfer Service (BITS)
BitLocker Drive Encryption
BitLocker Network Unlock
BranchCache
Client for NFS
Containers
Data Center Bridging
Direct Play
Enhanced Storage
Failover Clustering
Group Policy Management (Installed)
Host Guardian Hyper-V Support
I/O Quality of Service
IIS Hostable Web Core
Internet Printing Client
IP Address Management (IPAM) Server
LPR Port Monitor
Description
.NET Framework 3.5 combines the power of the .NET Framework 2.0 APIs with new technologies for building applications that offer appealing user interfaces, protect your customers' personal identity information, enable seamless and secure communication, and provide the ability to model a range of business processes.
< Previous
Next >
Install
Cancel

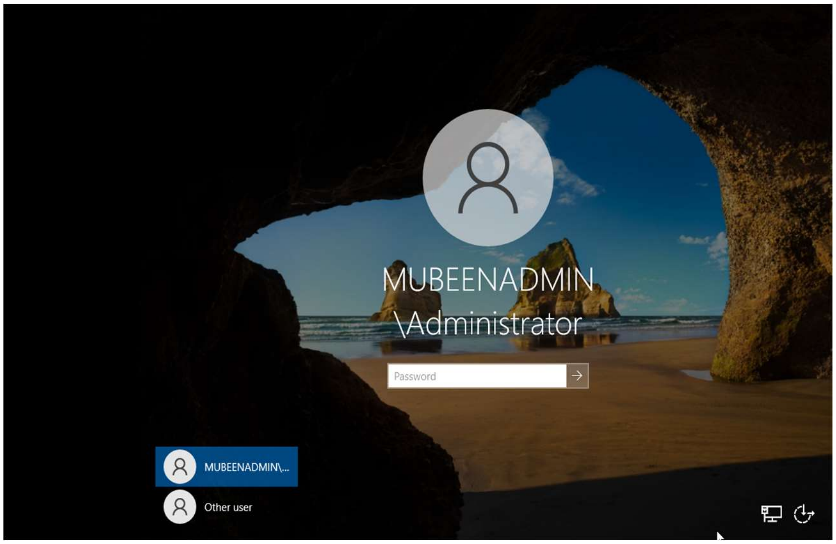

Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
WIN-P6TU2UVNTST
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Remote Access
Role Services
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
Fax Server
File and Storage Services (1 of 12 installed)
Host Guardian Service
Hyper-V
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Windows Deployment Services
Description
Remote Access provides seamless connectivity through DirectAccess, VPN, and Web Application Proxy. DirectAccess provides an Always On and Always Managed experience. RAS provides traditional VPN services, including site-to-site (branch-office or cloud-based) connectivity. Web Application Proxy enables the publishing of selected HTTP- and HTTPS-based applications from your corporate network to client devices outside of the corporate network. Routing provides traditional routing capabilities, including NAT and other connectivity options. RAS and Routing can be deployed in single-tenant or multi-tenant mode.



3.2 Promoting Active Directory Domain Services to Domain Controller



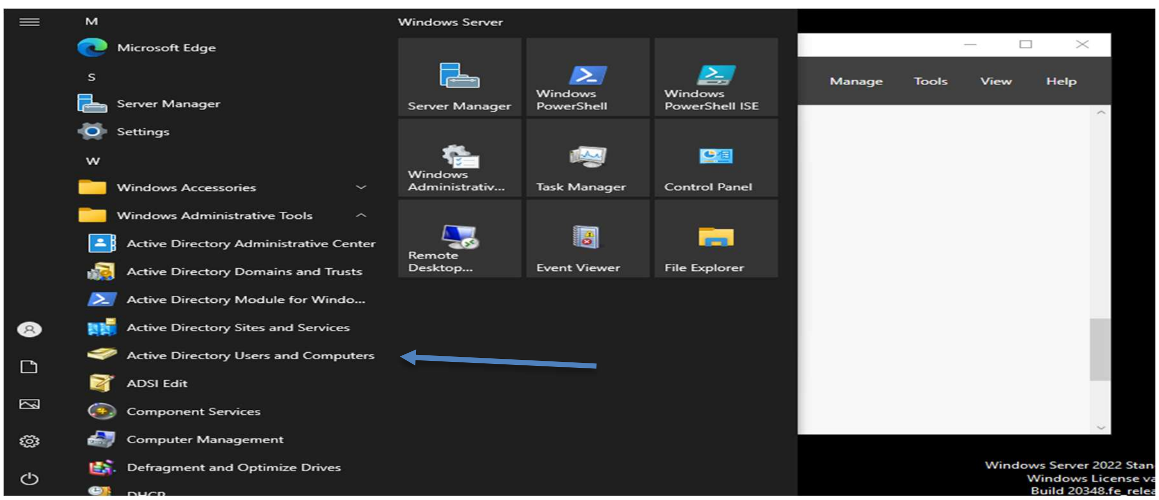


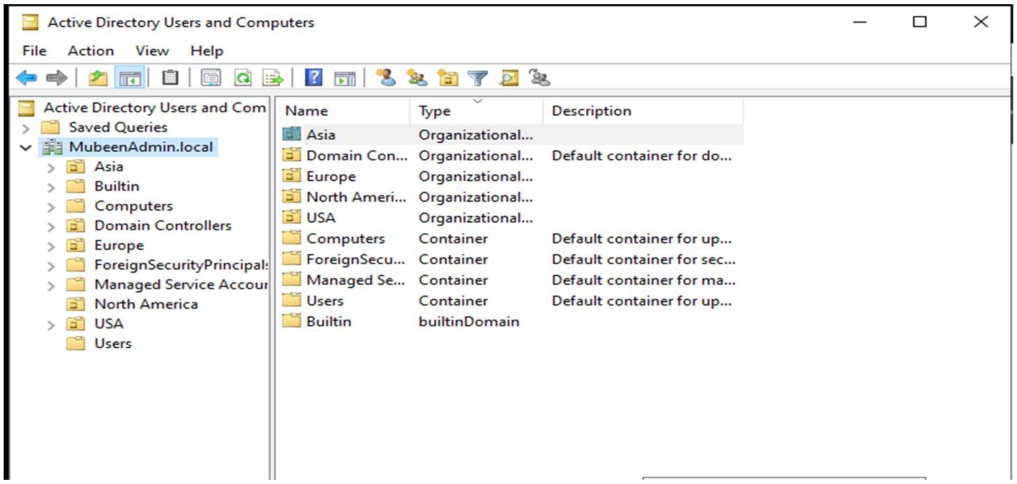


4. Domain Configuration

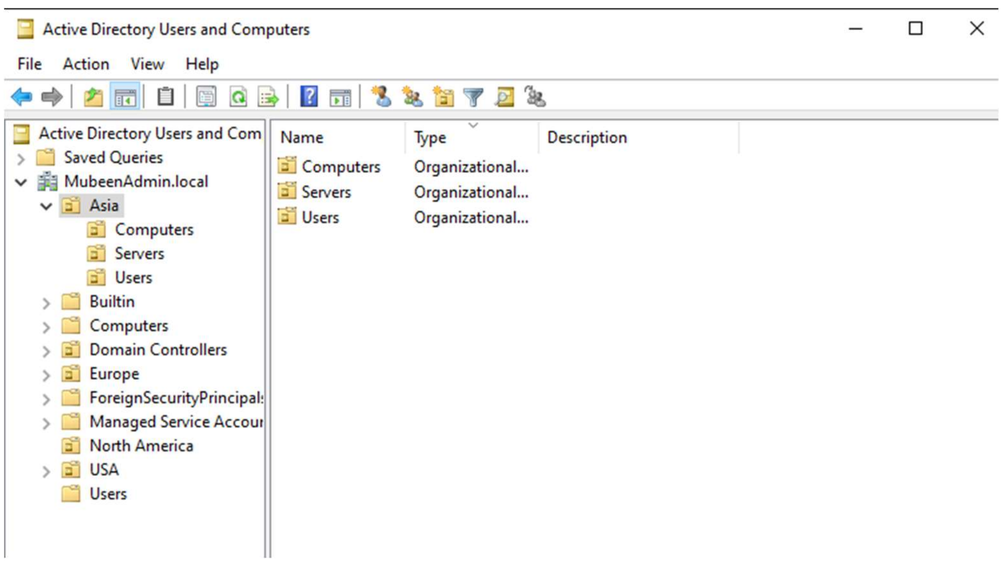
4.1 Adding Organizational Units (OUs)

- **Navigate to Active Directory Users and Computers**
- **Right click and create Organizational Units, I have already created 4 OUs (Asia, Europe, North America, and USA)**



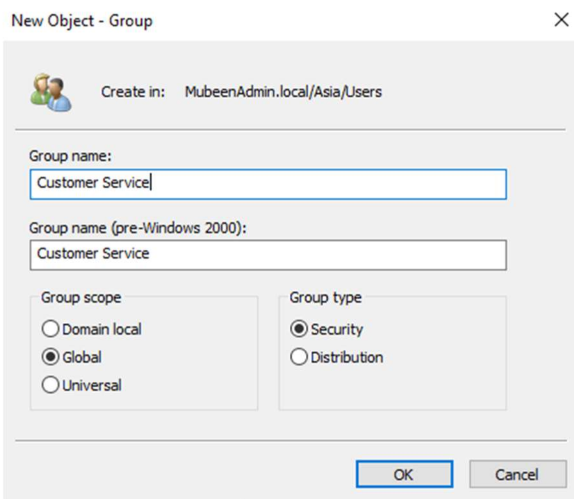


- **Inside the Organizational Unit (Asia, Europe, Noth America, and USA) we create another Organizational Unit (Server, Computers, Users)**

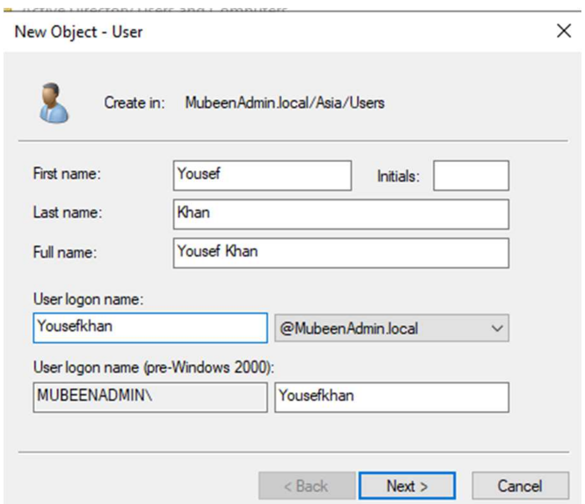


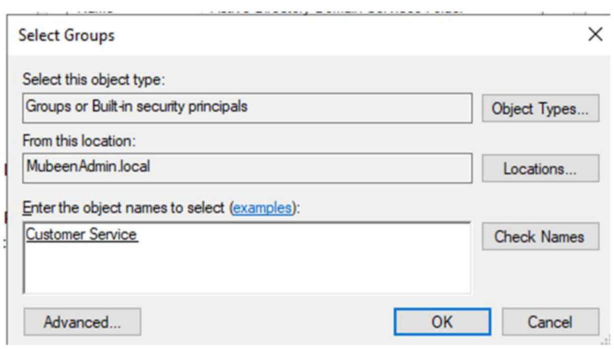
4.3 User and Group Creation

- **Inside the Organizational Unit we will create groups (Departments and Distribution List).**
- **Departments include: HR, IT, Finance, Sales, Customer Service**

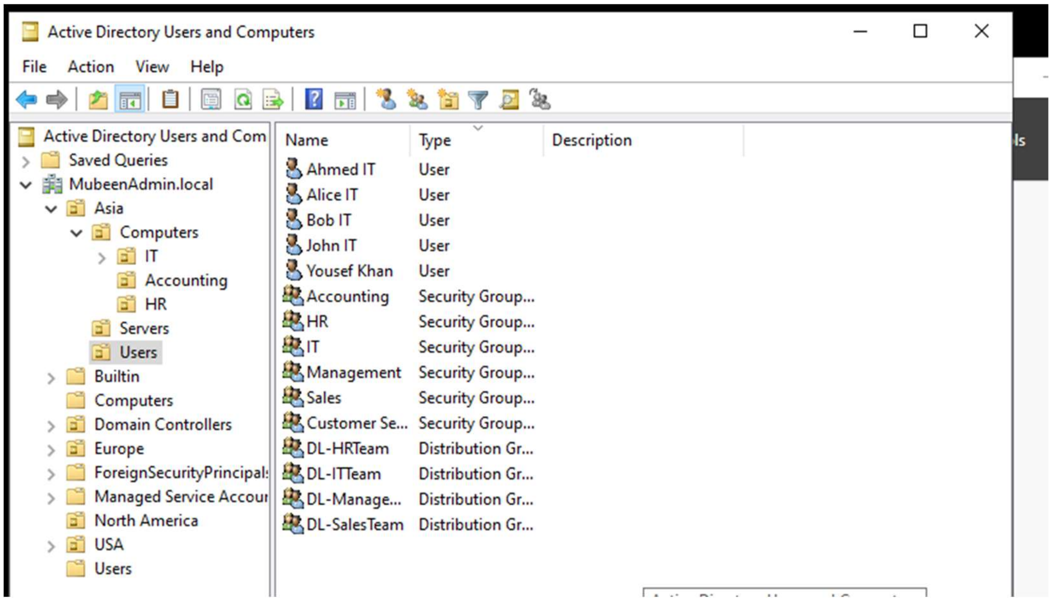


Now we will create User and add it to customer service department.

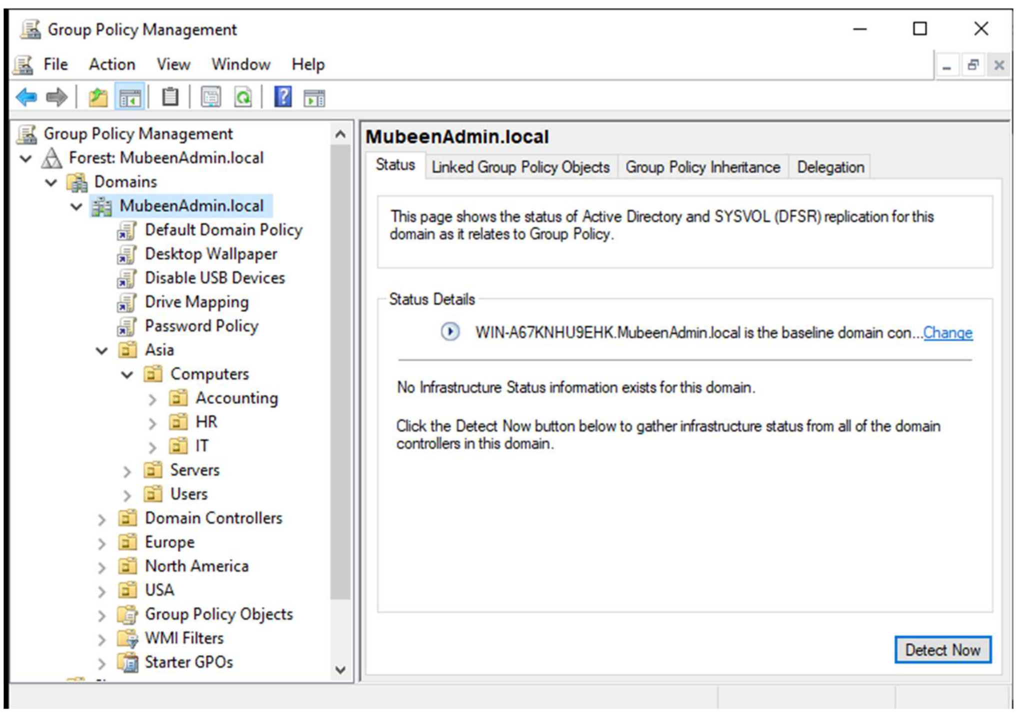




We will create Distribution list in similar way and overall our User will look like this:



5 Group Policy Management

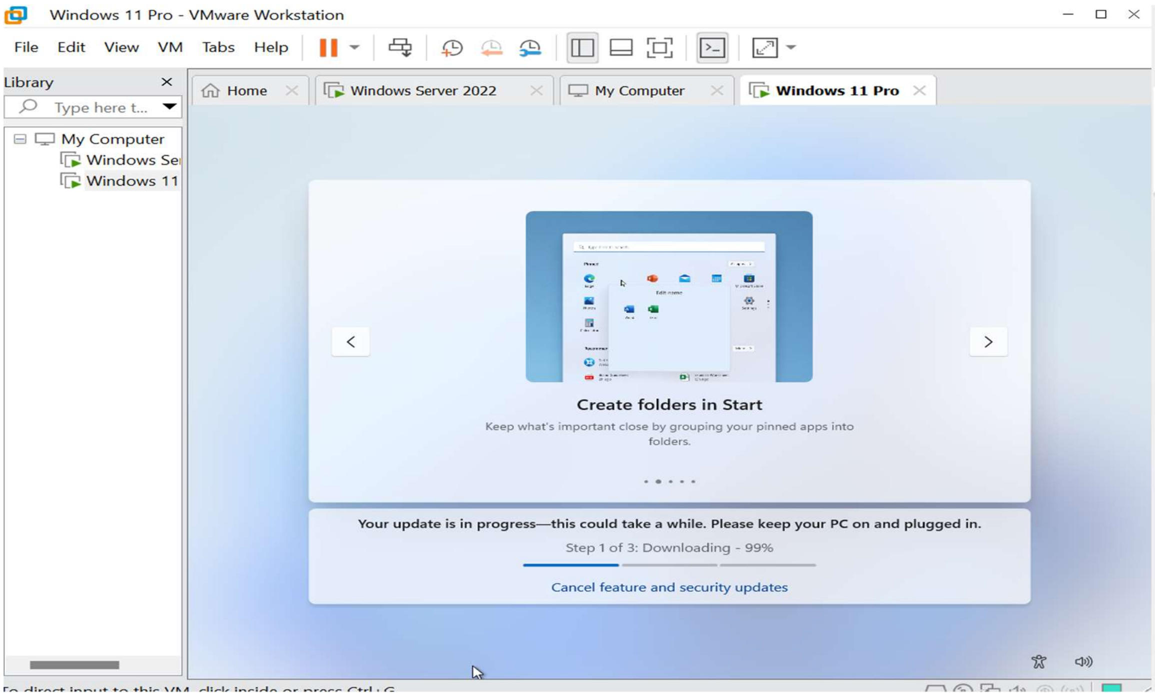


Created 5 Policies:

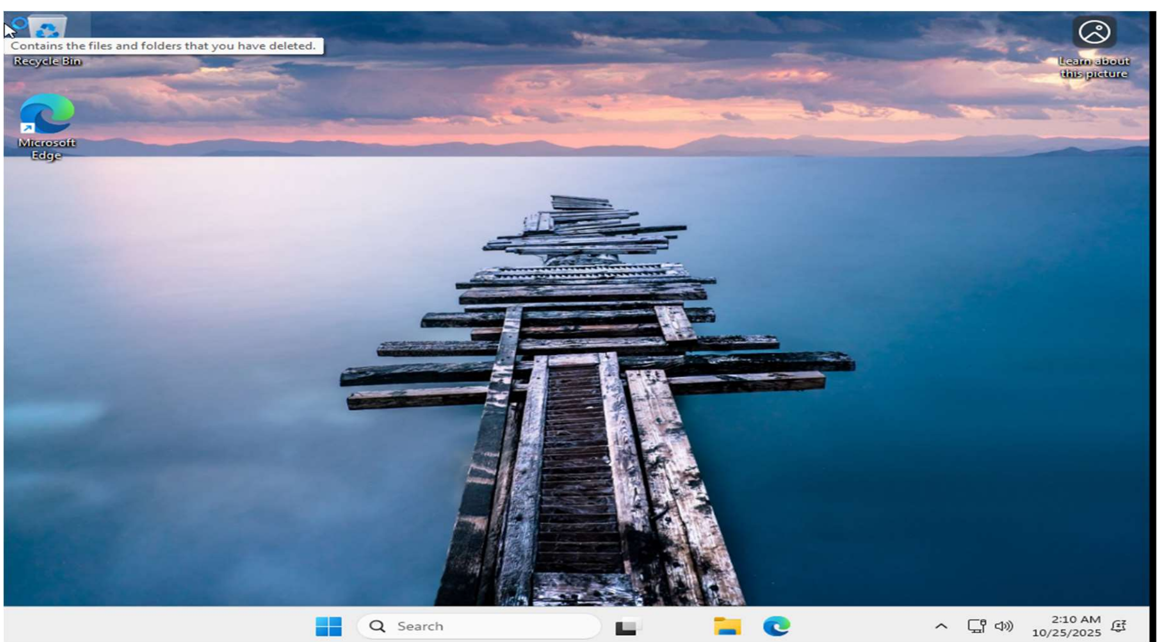
- **Desktop Wallpaper policy**
- **Disable USB Devices**
- **Drive Mapping**
- **Password Policy (Including Account Lockout policy)**

6. Joining Client Computers to the domain

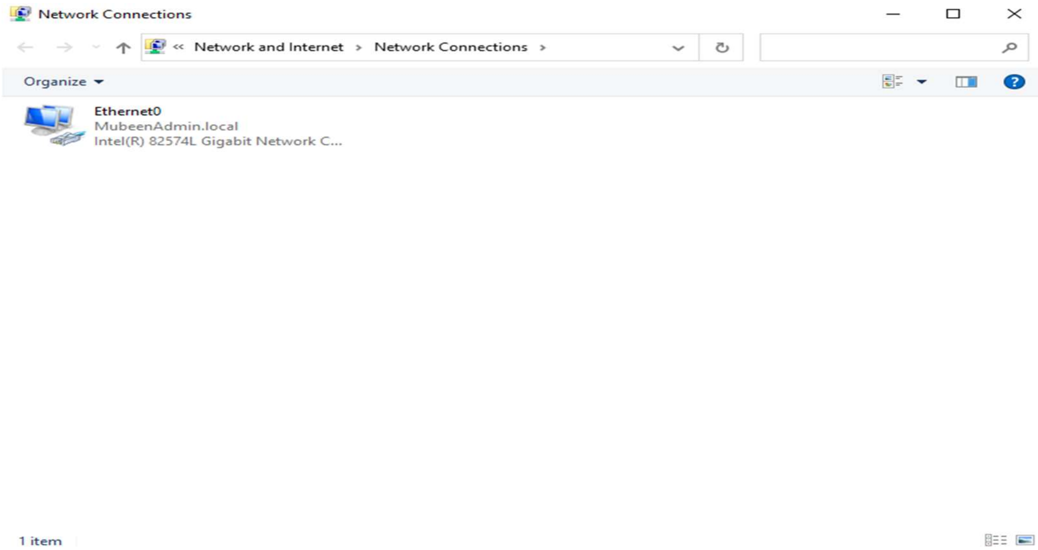
- **Installing Windows 11 Pro Iso and setting up in the VMware workstation pro.**

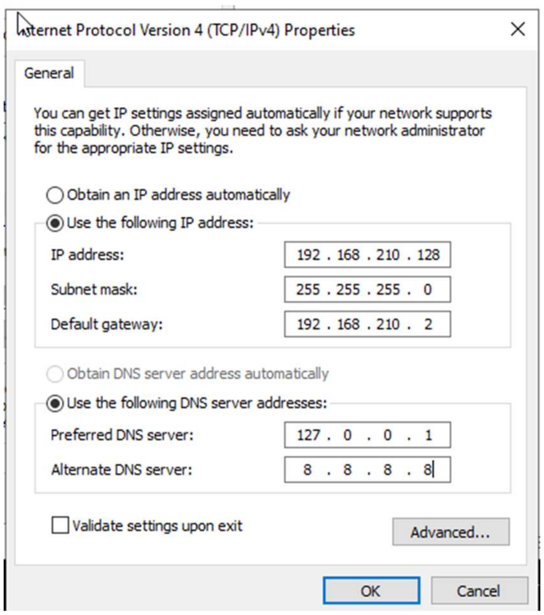


Choose "Domain Join Instead" option and setup the account.

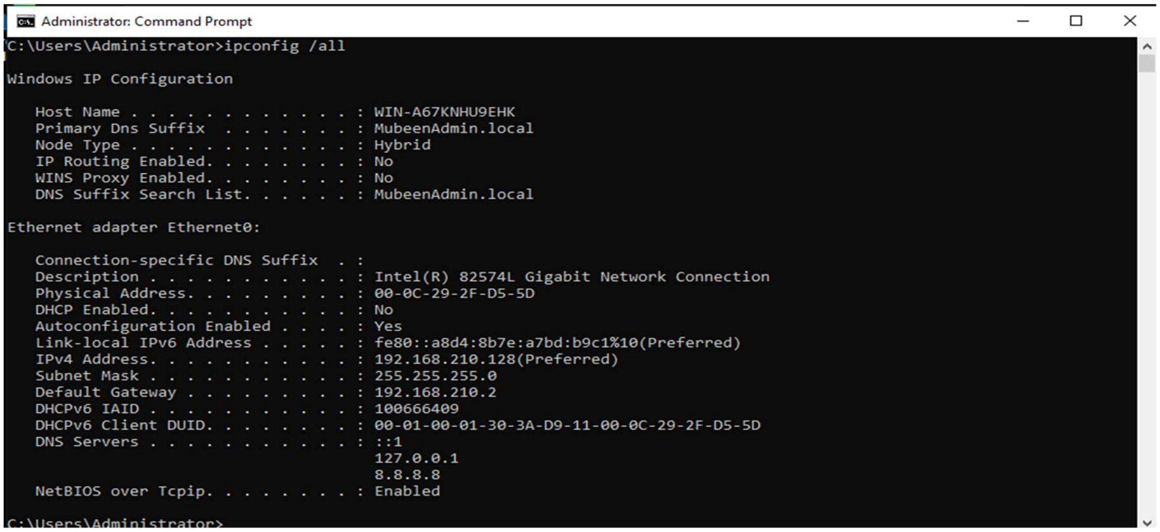


Configuring Static IP Address on Windows Server

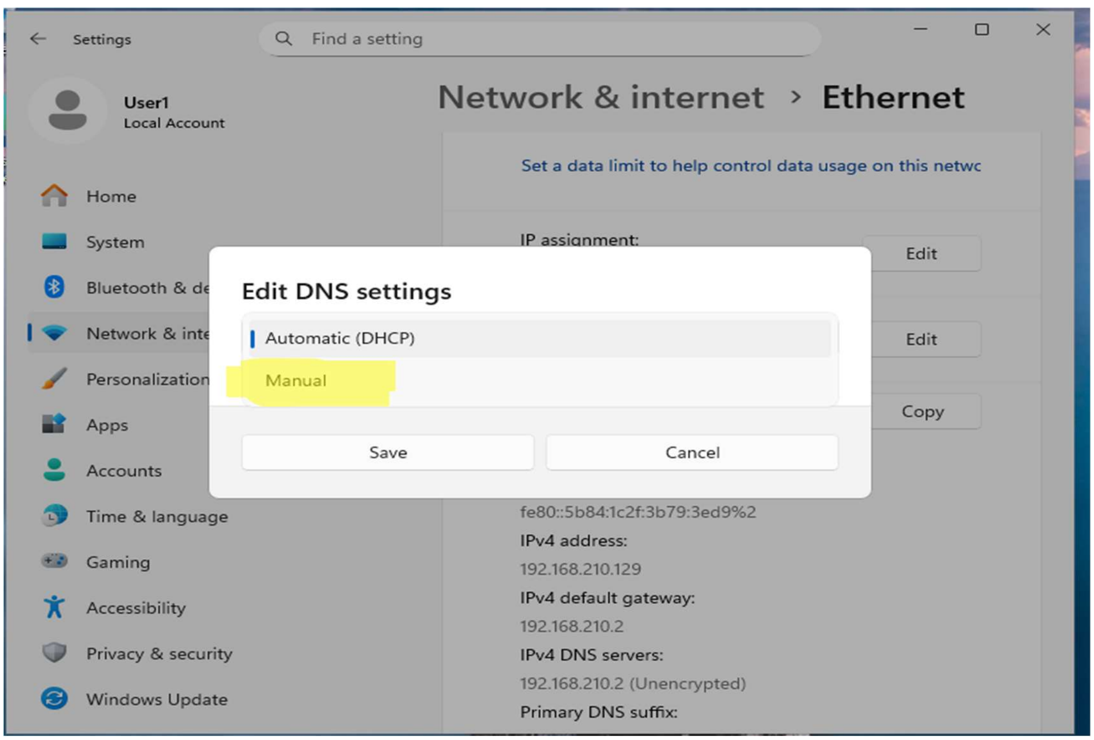


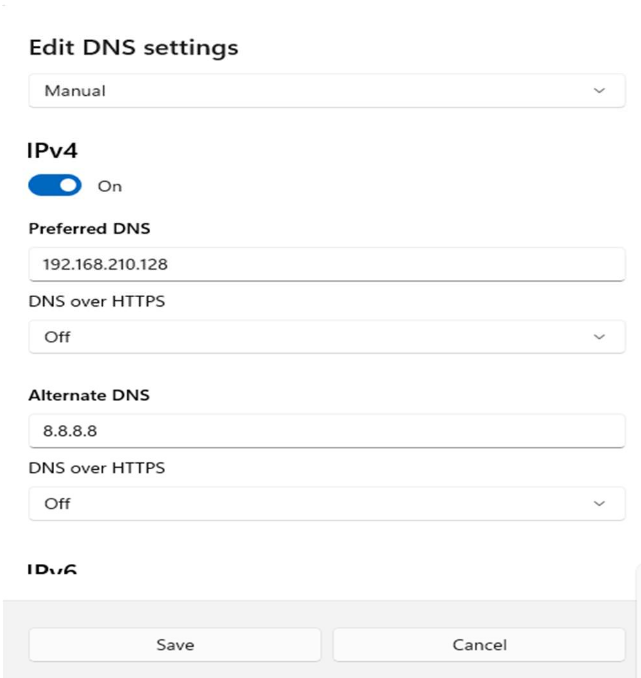


Verifying the changes of Static IP address using "ipconfig /all"

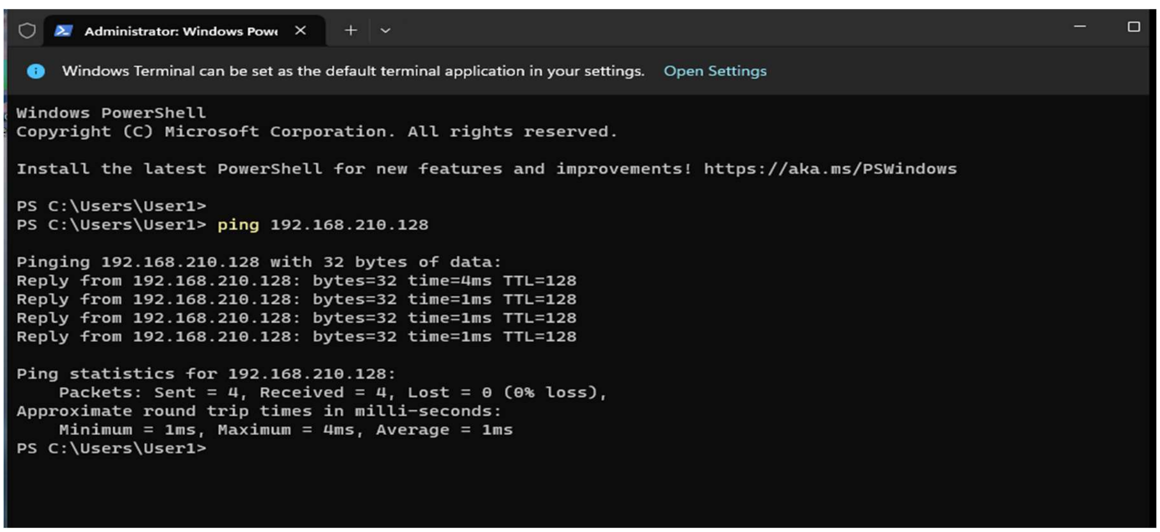


Setting DNS of Client to the Domain Controller

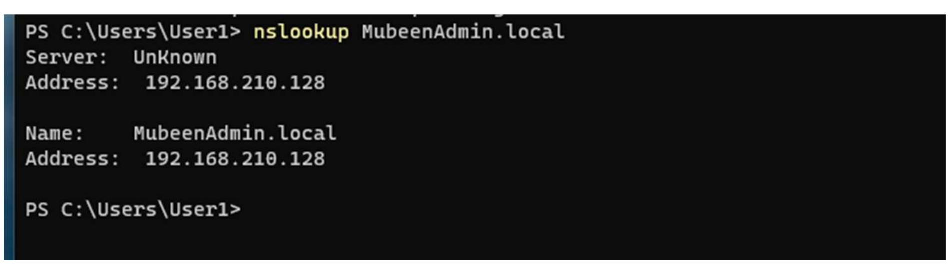




Then we ping it our domain and it should be pinging

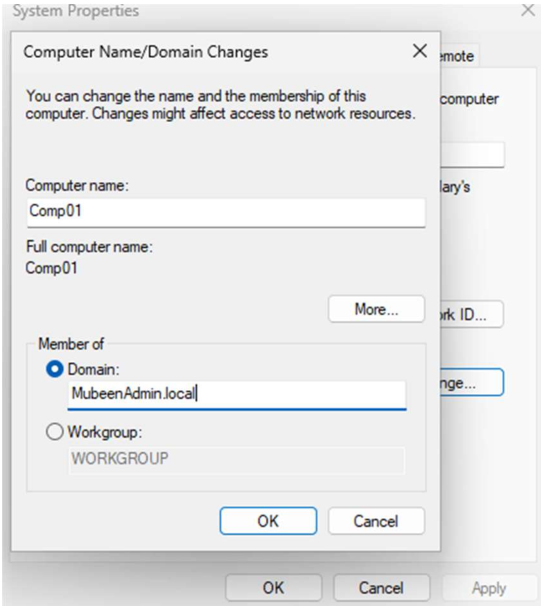


To check if the DNS is working correctly, we can use nslookup

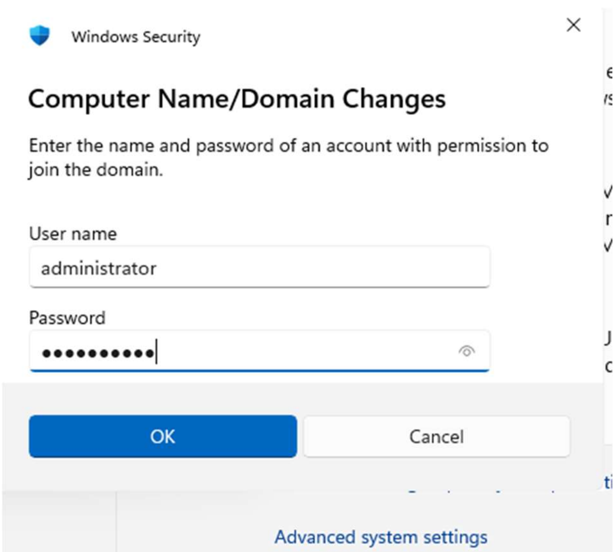


Joining the computer to domain

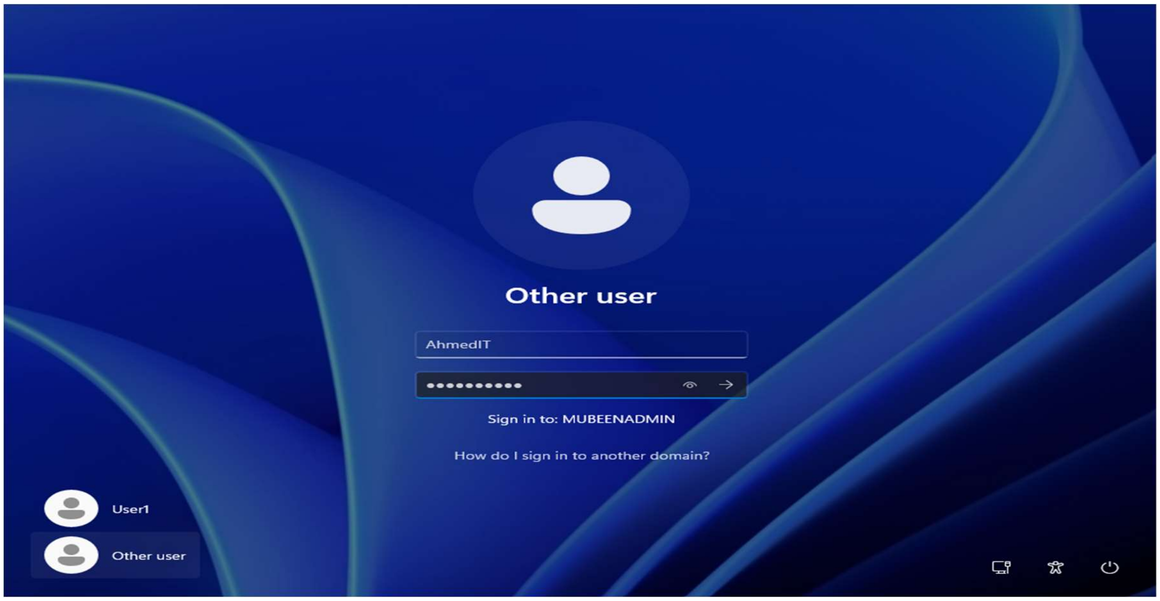
We rename the computer and enter our domain name



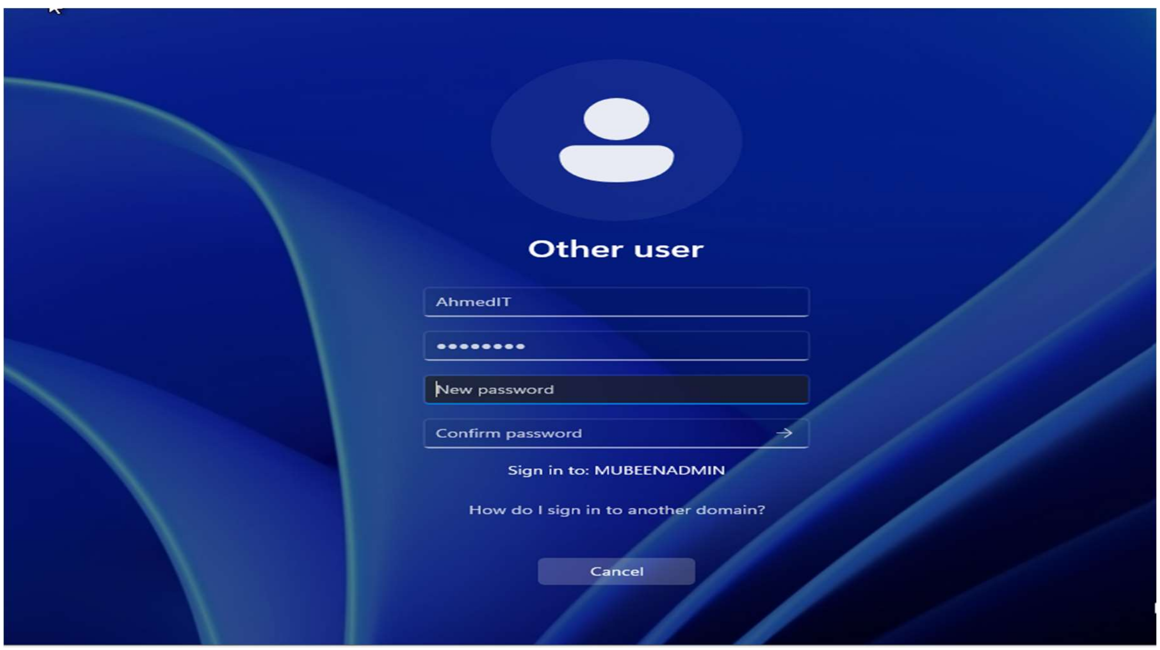
Enter the Admin credentials



We will try to login using one of the users we had created "Ahmed IT"

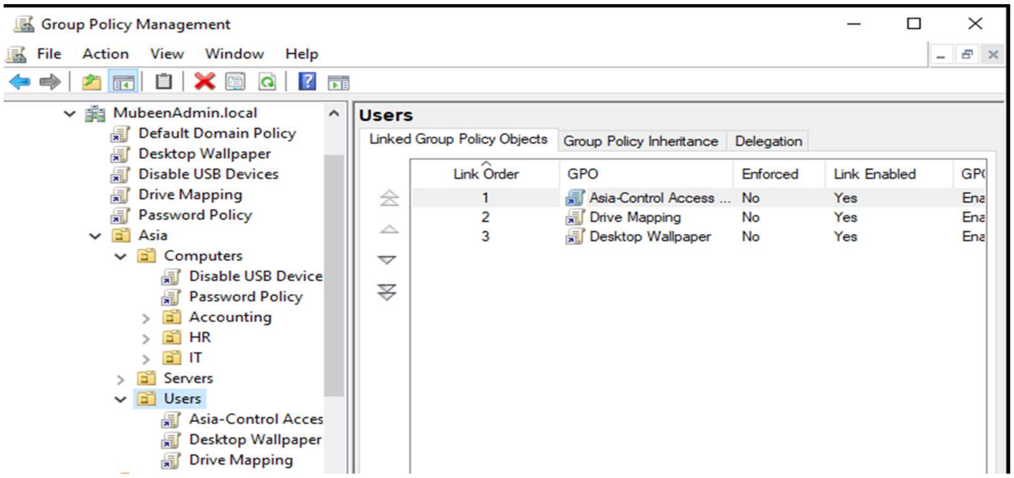


Since we chose “user password must be changed before login so we have to change the password:

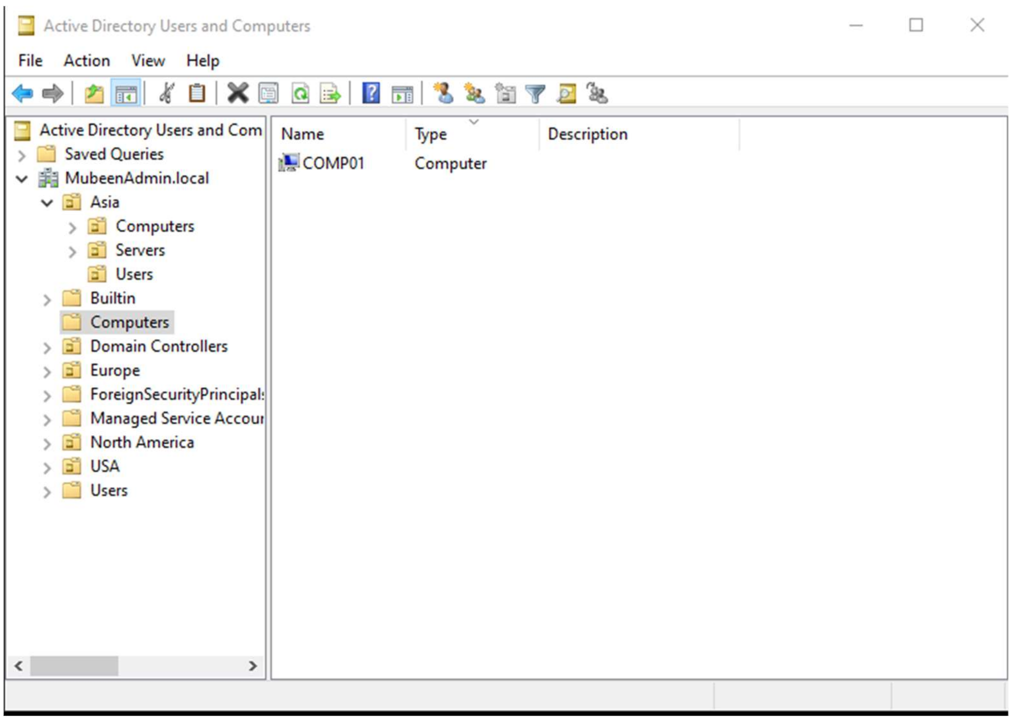


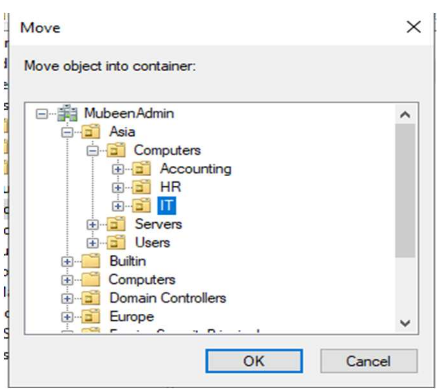
7 Testing and Verification

Now we will be implementing our GPOs, such as password policies under Computers, Desktop Wallpaper under Users and etc.

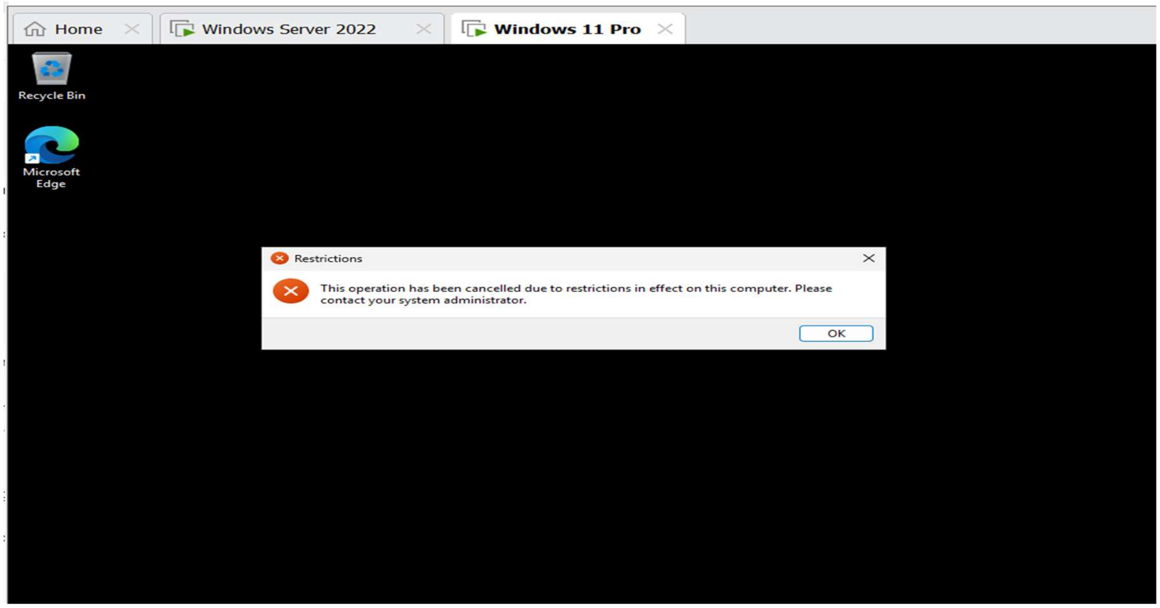


We also must move the Computer that we join to our domain to its respective OU

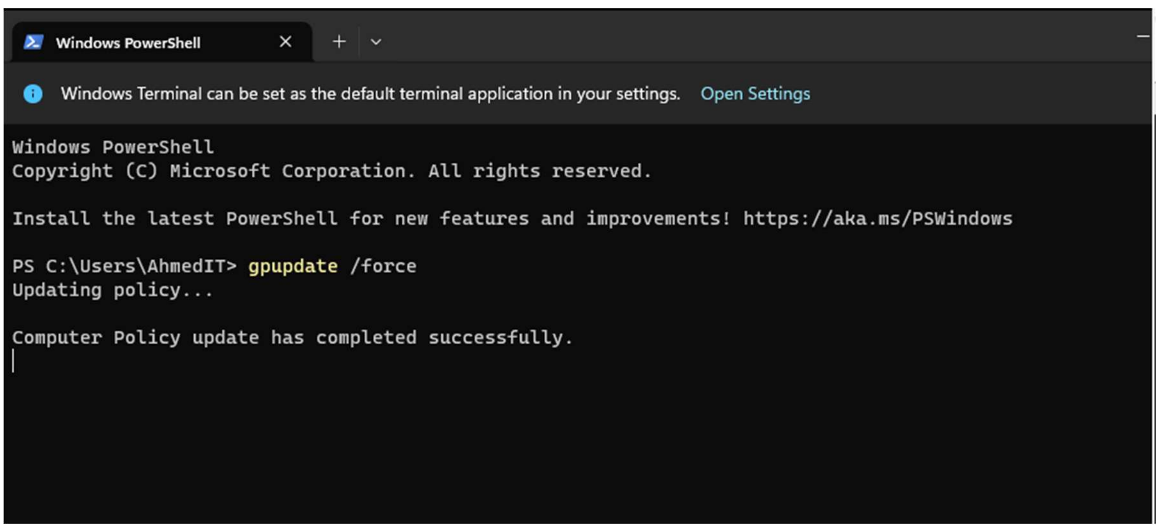




Now I tried accessing control panel on my client computer:



We could also force Group Policy update if it takes time using gpupdate /force



8 Conclusion and Recommendation

Project Summary

Successfully designed and deployed a full Windows Server 2022 domain infrastructure from scratch. Configured Active Directory with organizational units for different regions and departments, created user/group accounts, and implemented Group Policies for security (password policies, USB restrictions) and user environment management (drive mapping, desktop settings). Demonstrated end-to-end domain administration by joining Windows 11 clients to the domain and verifying all policies applied correctly.

Key Achievements:

- Built secure domain with structured AD organizational units
- Automated security and user settings via Group Policy
- Integrated client workstations into domain environment
- Established centralized enterprise management foundation

8.2 Recommendations for Future Enhancements

While the current setup provides a fully functional domain environment, several enhancements could be implemented to increase security, redundancy, and functionality in a production scenario:

1. **Implement a Redundant Domain Controller:** Deploy a second Domain Controller to provide fault tolerance and load balancing. This ensures authentication and directory services remain available if the primary server fails.
2. **Strengthen Security Policies:** Enhance GPOs to include more advanced settings, such as:
    - **Account Policies:** Implement a fine-grained password policy for administrative accounts.
    - **Audit Policies:** Enable auditing for logon events and object access to monitor for suspicious activity.
    - **User Rights Assignment:** Restrict local logon rights on client computers.
3. **Deploy Additional Roles and Services:**
    - **DHCP Server:** Automate IP address assignment to all clients on the network.
    - **DNS Server:** Further configure and secure the DNS role for reliable name resolution.
    - **File Services:** Set up centralized, permission-controlled shared folders for each department.
4. **Utilize Group Policy Preferences:** Further leverage GPO Preferences to deploy network printers, configure power plans, and manage registry settings across all domain-joined computers.
5. **Establish a Backup and Disaster Recovery Plan:** Implement a regular backup schedule for Active Directory and critical system state data to enable quick recovery from data corruption or hardware failure.